

Umais Yousaf

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Education

University of Colorado Boulder, MS in Computer Science - CGPA 4.0 Aug 2025 – May 2027

- **Coursework:** Transformers for Robotics, Neuro-Symbolic NLP, Distributed Systems

Lahore University of Management Sciences (LUMS), BS in Computer Science - CGPA 3.6 Aug 2019 – May 2023

- **Coursework:** NLP, DL, CV, ML, Data Mining, Stats & Data Analysis, Data Science, Applied Probability, HCI, SWE

Technologies

Frameworks/Tools: CUDA, HuggingFace, LangChain, LangGraph, Flask, React, Docker, MERN, Kafka, Spark, gRPC

Techniques: Vision Transformers, VQA, YOLO, OCR, RAG, Graph RAG, MCP, EDA, ETL / ETL

Languages: Python (PyTorch, TensorFlow, OpenCV), C++, JavaScript, SQL

Cloud: Azure (Azure ML, Azure Container Instances, Azure DevOps)

Experience

NSF Institute for Student-AI Teaming, Graduate Student Researcher - Boulder, CO · On Site Feb 2026 - Present

- Researching AI-assisted collaborative learning systems for classroom group activities.
- Investigating **RAG** pipelines including document chunking, embedding strategies, and retrieval methods.
- Integrating LLMs with real-time speech detection using **Whisper** to enable AI assistance during student collaboration.
- Conducting evaluation and iterative improvement of AI assistants deployed in real-world educational settings.

Reteta, Junior Consultant (AI Team) - Cranbury, NJ · Remote Nov 2024 - July 2025

- Performed data collection, cleaning, preparation, and **fine-tuned LLMs** to generate up to 95% accurate structured SOAP notes from patient-doctor interactions, reducing physician burnout by 75%.
- Achieved top-tier performance in medical NLP, enhancing the product's position in the healthcare AI market.
- Deployed and maintained a scalable ML backend on **Azure** with **Azure ML** and **Azure Container Instances**, and built CI/CD pipelines using **Azure DevOps** for automated testing, deployment, logging, and monitoring.
- Developed **NER**-based techniques to anonymize PII, ensuring HIPAA compliance.

Visionet-Systems, Junior Consultant (AI Team) - Lahore · On Site July 2023 - Nov 2024

- Developed a comprehensive PDF parsing pipeline and integrated with **Agentic AI and RAG** architectures, improving document retrieval efficiency by 20%. The pipeline detected and parsed text, tables, images, captions, headers, and footers.
- Used this pipeline to create **end-to-end Generative AI solutions** for the insurance industry, such as automated underwriting and claim management workflows to streamline business operations
- Built a clothing recommendation system and reverse image search using transformers, Siamese networks, and contrastive learning, enhancing recommendation accuracy by 15%.
- Created end-to-end RAG workflows using **LangChain** and **LangGraph**.
- Trained multimodal LLMs for VQA and image-based extraction, outperforming in-house OCR.
- Delivered live pre-sales demos to prospective clients, showcasing Generative AI solutions and building stakeholder confidence to accelerate deal progression.

Aletheia-AI (now BrickAndMortar.AI), ML and SWE Intern - Lahore · On Site Jun 2022 - Aug 2022

- Deployed real-time systems using YOLOv7 for vehicle, ANPR, and person detection, achieving >90% detection accuracy.
- Built computer vision pipelines for counting cement bags when loading onto trucks, and truck position verification on weighbridges, ensuring accurate weight measurement and significantly reducing manual intervention.
- Managed end-to-end data workflows: collection, annotation, training, and fine-tuning of models.

Projects

Real-Time Tabletop Perception Pipeline | Robotics/Perception Project - CU Boulder

- Built a real-time RGB-D perception pipeline in ROS 2 for tabletop object detection and localization for manipulation
- Integrated volumetric fusion and ESDF computation using NVBlox for collision-aware planning

LEGAL-UQA: Low-Resource Urdu-English Dataset | NLP/ML Project | [arXiv link](#)

- Created the first Urdu-English legal QA dataset from Pakistan's constitution using OCR extraction, manual refinement, and GPT-4-assisted translation

Lane Change Warning System | CV/ML Project

- Developed a dashcam video pipeline to detect vehicles (YOLOv7, ~96% mAP) and lanes (LaneNet, ~93% IoU)
- Used automatic dynamic homography to generate top-down view for lane-change detection in real time (25 FPS)